Under what conditions do outer giants coexist with inner small planets?

Orbital coherence, excitation history, and observational selection

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Abstract

Outer giants coexist with inner small planets when formation supplies the inner system and subsequent perturbations avoid destructive differential orbital excitation. We examine the dynamical part of this statement with a public-catalog audit, a screen of 1,719 synthetic architectures, 694 direct integration jobs including controls and numerical checks, and 1,008 reduced secular histories. In 24 deliberately stress-selected architectures, coherent eccentricity vectors yield 39/48 encounter-free realizations over 20,000 yr, compared with 11/48 circular starts; halving the timestep gives 39/48 and 15/48. Removing the giant yields 48/48, whereas shuffling inner apses at fixed eccentricity magnitudes yields 1/48. Selected coherent realizations persist for one million years, but others fail. Prescribed slow giant excitation and gradual removal of inner damping can generate low free eccentricity from circular starts. These histories are reduced disk models, and an apparent faster-growth survival disadvantage disappears on numerical refinement. Scattering produces both weakly forced survivors and damaging trajectories. A fixed California Legacy Survey selection contains four coexistence hosts among 58 detected-giant hosts. Their modeled coherent eccentricity harmonics are only 10^{-4}–2×10^{-2} m s^{-1}, making this sample a weak test of orbital coherence. Coexistence is therefore favored by weak differential forcing, sufficiently strong internal coupling or additional precession, and compatible eccentricity vectors populated by favorable histories. The calculations support this conditional explanation, but do not establish a universal giant-mass, eccentricity, or separation threshold, a population survival fraction, or a unique formation pathway.

Keywords: planetary systems — celestial mechanics — planet formation — methods: numerical — methods: statistical

1 Introduction

The presence of a giant planet outside a compact system does not by itself determine the fate of the inner planets. Coexistence requires at least two physical conditions: the inner planets must form, and the later system must avoid destructive evolution. A third condition governs what is observed: both populations must be detectable. These conditions depend on different variables and should not be inferred from a single observed correlation.

Formation calculations show that the timing of inner growth, the supply of solids, and disk thermal structure affect whether small planets develop alongside cold giants [1]. An exterior giant can restrict inward material transport, but its presence does not imply that the inner region was empty when that restriction began. The present work does not calculate embryo growth or pebble transport. We address the subsequent dynamical requirement, while keeping this formation condition explicit.

The observed association between inner small planets and outer giants also depends on population definitions. Positive associations have been reported [2, 3, 4], with differences involving giant mass [5], metallicity conditioning [6], and the distinction between super-Earths and sub-Neptunes [7]. The hot-planet sample of Liu et al. [8] and the warm-planet sample of Ruggieri et al. [9] probe different regions of this problem. A single occurrence enhancement is not a universal coexistence condition.

The relevant dynamical ingredients are well established. An inner system can resist a companion when its mutual coupling dominates differential external forcing, while secular resonances can amplify the response [10]. Rapid relativistic precession can suppress eccentricity excitation at very small orbital radii, including in inclined systems [11]. At fixed final outer architecture, the history of scattering changes the excitation inherited by the inner system [12]. The instantaneous masses and orbital sizes therefore leave out physically consequential information.

An additional distinction is between scalar eccentricity and the eccentricity *vector*. Circular inner orbits in the presence of an eccentric companion generally contain free secular oscillations. A coherently precessing eccentric configuration can have less damaging relative motion even though its individual eccentricities are larger. Forced-eccentricity initial conditions improve packing stability in a stellar-binary setting [13], and aligned eccentric configurations can survive longer in closely spaced planetary systems [14]. Smooth external excitation and damping can populate coherent states [15, 16]; these are precedents, not mechanisms introduced here.

We combine three tests. First, we compare controlled inner states at fixed giant properties to isolate dynamical sensitivity. Second, we ask whether initially circular systems reach low free eccentricity through prescribed

growth, damping removal, or self-consistent giant scattering. Third, we use public survey products to assess whether the predicted orbital states can presently be distinguished. The resulting answer is conditional: weak forcing can permit coexistence without special coherence, whereas coherence may matter near a more strongly perturbed boundary. We quantify the evidence and its numerical limitations rather than fitting a universal boundary.

2 A conditional dynamical framework

2.1 Architecture, state, and history

Let $\mathcal{A}$ describe the full architecture, including inner spacing, planet masses, mutual inclinations, and additional companions. Let $\boldsymbol{z}$ denote the eccentricity vectors and h their excitation history. A survival probability at age t would require averaging a conditional survival function,

$$P_{\rm surv}(t \mid \mathcal{O}) = \int S(t \mid \mathcal{A}, \boldsymbol{z}, h)\, p(\mathcal{A}, \boldsymbol{z}, h \mid \mathcal{O})\, d\mathcal{A}\, d\boldsymbol{z}\, dh, \tag{1}$$

where $\mathcal{O}$ denotes the available observations. This expression is a bookkeeping statement, not an inferred model. We calculate selected values of S over finite horizons. We do not infer the distribution supplied by formation, deproject masses, or estimate the posterior average in this equation.

A large giant pericenter relative to the outermost small planet often reduces external perturbations. Strong inner mutual coupling can reduce differential excitation, and additional apsidal precession can move a system away from a secular resonance. None supplies a universal one-parameter criterion: the response also depends on the spectrum of precession frequencies, eccentricity vectors, resonance proximity, inclination, and perturbation history. Stable coexistence need not be coplanar or apsidally aligned. Our principal numerical comparison isolates the planar eccentricity channel.

2.2 Secular states and free-mode action

For planar orbits, write $z_j = e_j \exp(\mathrm{i}\varpi_j)$. In linear Laplace–Lagrange theory,

$$\dot{\boldsymbol{z}} = \mathrm{i} A \boldsymbol{z}. \tag{2}$$

We retain the giant as a gravitating member, including its backreaction. For $j \neq k$, define α_{jk} as the smaller-to-larger semimajor-axis ratio and $\bar{\alpha}_{jk} = \alpha_{jk}$ if j is interior, otherwise unity. With $n_j = [G(M_* + m_j)/a_j^3]^{1/2}$,

$$A_{jj} = \frac{n_j}{4} \sum_{k \neq j} \frac{m_k}{M_* + m_j} \alpha_{jk} \bar{\alpha}_{jk} b^{(1)}_{3/2}(\alpha_{jk}), \tag{3}$$

$$A_{jk} = -\frac{n_j}{4} \frac{m_k}{M_* + m_j} \alpha_{jk} \bar{\alpha}_{jk} b^{(2)}_{3/2}(\alpha_{jk}). \tag{4}$$

The Laplace coefficients are evaluated through a hypergeometric representation and checked independently by quadrature.

With $W_{jj} = [m_j \sqrt{(M_* + m_j) a_j}]^{1/2}$, the adopted matrix is symmetrized as $H = WAW^{-1}$. Let U be its orthonormal eigenvectors, g_k the frequencies, and $V = W^{-1}U$. The mode k_g with maximum U^2_{gk} is our giant-dominated reference. Normalized to the initial giant eccentricity, the prepared and circular states are

$$\boldsymbol{z}_{\rm coh} = e_g \boldsymbol{V}_{k_g} / V_{g k_g}, \tag{5}$$

$$\boldsymbol{z}_{\rm cold} = (0, \ldots, 0, e_g)^{\mathsf{T}}. \tag{6}$$

All 48 selected reference modes have aligned apses. This does not assert that alignment is the only protective configuration. Nor is a linear eigenmode an exact periodic solution of the finite-eccentricity gravitational problem.

Using $\boldsymbol{c} = U^{\mathsf{T}} W \boldsymbol{z}$, we normalize the non-reference action as

$$\mathcal{F} = \frac{\sum_{k \neq k_g} |c_k|^2}{\sum_{k \neq k_g} |c_{k,\rm cold}|^2}, \qquad \Delta g = \min_{k \neq k_g} |g_k - g_{k_g}|, \tag{7}$$

where the cold state has the same instantaneous giant vector. Thus $\mathcal{F} = 0$ for the pure reference mode and $\mathcal{F} = 1$ for circular inner orbits. For $\boldsymbol{z} = (1-\eta)\boldsymbol{z}_{\rm coh} + \eta \boldsymbol{z}_{\rm cold}$, $\mathcal{F} = \eta^2$ exactly. This normalization is a diagnostic, not a new invariant or a universal survival threshold. When the giant eccentricity is tiny, the denominator can produce a large ratio despite weak absolute excitation. Nonlinear evolution, inclination, semimajor-axis drift, and osculating short-period terms further limit its interpretation.

2.3 Relative geometry and encounter endpoints

For two coplanar ellipses with $p_j = a_j(1-e_j^2)$ and $p_1 \neq p_2$, equality of the polar radii gives the instantaneous intersection diagnostic

$$X_{12} = \frac{|p_2 z_1 - p_1 z_2|}{|p_2 - p_1|}. \tag{8}$$

The osculating curves intersect or touch when $X_{12} \geq 1$. By contrast, $(a_1 e_1 + a_2 e_2)/(a_2 - a_1) \geq 1$ only indicates overlap of radial ranges. Alignment can reduce the vector difference in equation (8) while those ranges overlap.

Neither diagnostic guarantees dynamical stability or predicts encounter time. Mean-motion resonance overlap can matter before orbit crossing [17]; angular-momentum-deficit stability supplies a useful sufficient secular condition, but failure of that condition does not prove instability [18]. We therefore use secular geometry for selection and full gravitational integration for finite-time endpoints.

3 Data and numerical design

3.1 Public catalog context

The retained NASA Exoplanet Archive default-solution snapshot [19], retrieved on 14 October 2026, contains 6,445 planets around 4,842 named hosts. For this contextual audit, an inner small planet has $a < 1$ au and either radius 1–4 $R_\oplus$ or measured/minimum mass 1–20 $M_\oplus$. A giant has measured/minimum mass 0.3–13 M_J and $a = 1$–10 au. Limits are excluded from the relevant selection branch; missing semimajor axes are inferred from period and stellar mass when possible.

This definition gives 37 coexistence hosts, 42 qualifying giants, and ten hosts with multiple selected small planets. Of the giants, five lack eccentricities and two have eccentricity upper limits. Among the 35 uncensored values, the giant-weighted median $q_g/a_{i,\mathrm{out}}$ is 17.78, or 6.57 in the multiple-small-planet subset. In comparison, our stress-selected simulations span 1.81–2.87, with median 1.95. They deliberately probe a more strongly perturbed regime than most catalog examples.

The count changes with the definition: giant lower-mass cuts of 0.2, 0.3, and 0.5 M_J give 38, 37, and 33 hosts. At 0.3 M_J, radius-only and mass-only definitions give 24 and 32. The catalog lacks a common survey denominator and recovery function. These are detected examples, not an occurrence measurement or observationally fitted simulation initial conditions.

3.2 Controlled architecture ensemble

We generate 1,800 proposals around a solar-mass star, with random seed 20261014. Each has two or three coplanar inner planets, $a_1 = 0.2$ au, common adjacent semimajor-axis ratio uniform on 1.22–1.65, and independently log-uniform masses of 1–12 $M_\oplus$. The giant has log-uniform $a_g = 1$–4 au and mass 0.3–10 M_J, and uniform $e_g = 0.08$–0.50. Rejecting $q_g/a_{i,\mathrm{out}} < 1.8$ and prepared inner eccentricities above 0.5 leaves 1,719 architectures.

We sample the analytic secular evolution at 4,096 epochs over 10^5 yr. The 24 highest circular-start intersection diagnostics form the stress cohort. A further 24 architectures are drawn from the remaining pool with seed 2718 as a comparison cohort. Selection on a circular-start diagnostic intentionally enriches the first cohort for possible contrasts; the comparison is not an unbiased population survival experiment or a held-out sample.

Each architecture has four inner-state prescriptions: circular; prepared coherent; prepared with independent 10% Gaussian fractional eccentricity perturbations and 10° Gaussian apsidal perturbations; and prepared eccentricity magnitudes with uniformly shuffled apses. The giant starts at $\varpi_g = 0$. Mean anomalies are randomized with seeds 101 and 202 and matched across prescriptions. This gives 384 primary integrations.

We use REBOUND 5.2.2 with WHFast [20], timestep $P_1/40$, corrector order 11, and a 20,000-yr horizon. Initial elements are heliocentric and subsequently transformed to the center of mass. Encounter termination uses one fixed distance: the smallest initial mutual Hill radius among adjacent inner pairs, applied to all particle pairs in that architecture. Escape beyond 100 au, unbound osculating orbits, and sampled pericenters inside 0.00465 au are also monitored. Every non-completed primary run reaches the encounter endpoint. An encounter is not a simulated destruction event; remaining multiplicity is not followed afterward.

Additional runs remove the giant, halve the timestep, compare with adaptive IAS15 [21], vary the initial mode loading, and extend selected systems to 10^6 yr. The baseline comprises 628 distinct jobs. The history study adds 66, for 694 direct jobs in total. This count includes pilots, controls, and reruns of the same initial configurations, not 694 independent systems (Appendix A).

Table 1: Encounter-free completions at 20,000 yr. Counts are conditional on the selected architectures.

Initial inner state	Stress cohort	Comparison
Circular	11/48	48/48
Coherent	39/48	48/48
Coherent, perturbed	37/48	48/48
Apses shuffled	1/48	44/48
Circular, no giant	48/48	—
Circular, $P_1/80$	15/48	—
Coherent, $P_1/80$	39/48	—

4 Dependence on orbital state

4.1 Prepared, circular, and shuffled states

Table 1 gives the primary endpoint counts. Among 48 stress-cohort realizations per prescription, 39 coherent states complete the horizon, versus 11 circular states. Mildly perturbed coherent states yield 37 completions; shuffled apses yield only one. In the comparison cohort, circular, prepared, and mildly perturbed states all give 48/48, while shuffled states give 44/48.

The shuffled control preserves scalar eccentricity magnitudes and changes apsidal phases and mode content. Only two stress-cohort shuffles initially have intersecting inner conics; excluding them leaves 1/46 completions. The poor outcome is therefore not explained by initializing most of those systems with intersecting ellipses. Conversely, removing the giant from all 48 matched circular stress realizations gives 48/48 completions. The particular inner systems do not encounter in isolation over the test interval.

The coherent advantage remains in aggregate at half the timestep: 39/48 coherent and 15/48 circular completions. However, 12 of 96 matched endpoint labels differ between resolutions. The totals must not be interpreted

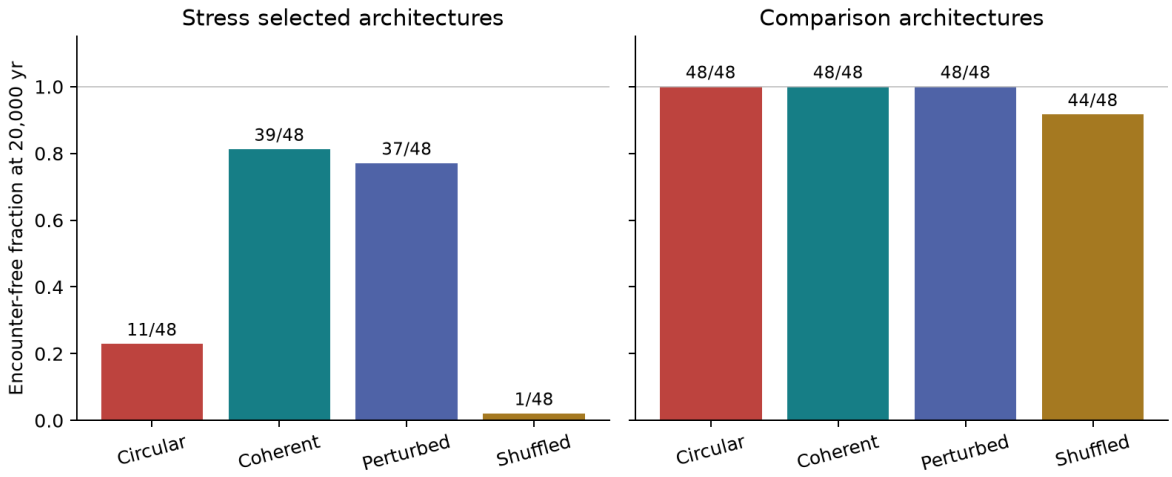


Figure 1: Primary controlled experiment. The stress cohort was selected for large circular-start secular intersection diagnostics; the comparison cohort was randomly selected from the remaining pool. Two mean-anomaly realizations per architecture are matched across states. These completion fractions describe selected finite-time integrations, not population occurrence or long-term survival probabilities.

as exact probabilities. Independent IAS15 and finer WHFast runs agree on the selected cases 24 and 1332, where circular starts encounter and prepared starts complete 20,000 yr. Case 916 is a failure example: both states encounter with both validation methods, including some configurations that completed at the coarser resolution.

4.2 Continuous mode loading

Case 24 has inner semimajor axes of 0.2000, 0.300735, and 0.452207 au. The corresponding masses are 1.0104, 8.0459, and 11.6860 $M_\oplus$. Its 6.3436 M_J giant lies at 1.626068 au. At the sampled $e_g = 0.409649$, $q_g/a_{i,\mathrm{out}} \simeq 2.12$.

Holding the masses and axes fixed, we scan seven giant eccentricities from 0.15 to 0.45 and five loadings $\eta = 0, 0.25, 0.5, 0.75, 1$, each with two matched phases. The initial inner vectors are $(1-\eta)\boldsymbol{z}_{\mathrm{coh}}$ and the giant vector is unchanged, so $\mathcal{F} = \eta^2$ initially. All 42 realizations with $\eta \leq 0.5$ complete. Circular states complete through $e_g = 0.30$ but encounter at each sampled value from 0.35 to 0.45. At $\eta = 0.75$, both phases complete through 0.35, one at 0.40, and neither at 0.45 (Fig. 2). This is a resolved dependence on initial state in one architecture, not a calibrated universal eccentricity cutoff.

4.3 Persistence and delayed failure

Twelve integrations of cases 24 and 1332 use $P_1/80$ and a 10^6-yr horizon. Both coherent and both perturbed case-24 realizations complete, while its circular starts encounter by 4,451 yr. For case 1332, one coherent realization completes and one encounters at 199,309 yr; the perturbed realizations encounter at 21,718 and 9,333 yr, and both circular starts before 438 yr. Across the two selected

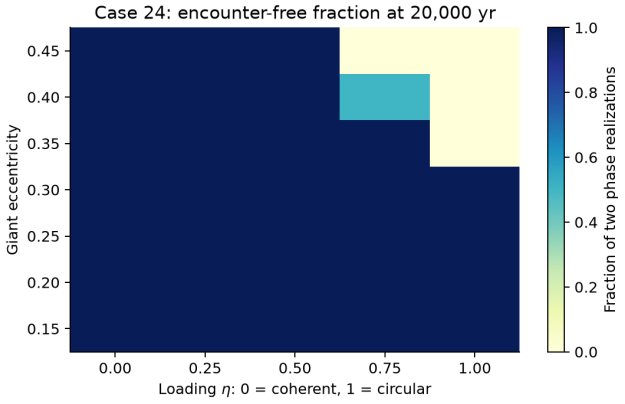


Figure 2: Case-24 loading scan. The prepared mode has $\eta = 0$, while circular inner orbits have $\eta = 1$. Each cell contains only two phase realizations. Changing η changes mode content and angular momentum deficit; it does not specify a formation timescale.

architectures this gives 3/4 coherent, 2/4 perturbed, and 0/4 circular completions.

These extensions demonstrate that an advantage can persist for one million years, while a 20,000-yr completion can conceal delayed failure. They do not yield a lifetime distribution or gigayear stability certificate. Linear mode preparation does not remove nonlinear interactions, resonances, or sensitivity to phase.

5 Generating coherent orbital states

Prepared initial conditions establish a possible dynamical advantage, but do not show whether formation populates

those states. We therefore begin the inner system circular or nearly circular and evolve prescribed giant excitation, damping removal, or giant instabilities. The disk calculations model orbital responses, not self-consistent planet formation.

5.1 Linear excitation and gas removal

For each of the 48 selected architectures, we solve

$$\dot{\boldsymbol{z}} = [\mathrm{i}A + \mathrm{diag}(-\gamma_i, \ldots, -\gamma_i, +\gamma_g)]\, \boldsymbol{z}, \tag{9}$$

starting at zero inner eccentricity and $e_g = 0.001$. Here $\gamma_g = 1/\tau_g$ and $\gamma_i = d/\tau_g$. We evaluate the matrix exponential and stop when the giant reaches the specified architecture's target eccentricity. Five values $x = \tau_g \Delta g = 0.01, 0.1, 1, 10, 100$ and three damping ratios $d = 0, 1, 10$ give 720 histories. The inverse angular-frequency gap $1/\Delta g$ is not a full 2π secular period.

Without inner damping, median final $\mathcal{F}$ decreases from 0.99985 at $x = 0.01$ to 0.43766 at $x = 1$, 0.00835 at $x = 10$, and 7.2×10^{-5} at $x = 100$. Slow forcing allows the inner vectors to approach the evolving coherent response. Strong damping changes that response: at $x = 10, d = 10$, median $\mathcal{F} = 0.48458$. Damping can hold the system away from the gas-free mode, so immediate removal need not leave low free eccentricity.

From all 48 endpoints at $x = 10, d = 10$, we switch off giant pumping and set $\gamma_i(t) = \gamma_{i,0} \exp(-t/\tau_d)$. Six removal parameters $y = \tau_d \Delta g = 0, 0.01, 0.1, 1, 10, 100$ give 288 further histories. We follow nonzero removal times for $12\tau_d$. Median $\mathcal{F}$ is 0.48458 for instantaneous removal, 0.47833 at $y = 0.1$, 0.15345 at $y = 1$, and about 2×10^{-11} at $y = 10$. The last value reflects the ideal linear model and finite residual damping at the stopping time; it is not a physically meaningful ten-decimal-place coherence prediction.

The mechanisms have close precedents in gradual excitation and apsidal capture [15], and in the response to growth, damping, and scattering [16]. Hydrodynamic calculations permit giant eccentricity growth in some disk cavities [22], motivating the sign of the prescribed forcing. They do not calibrate our exponential rates or constant damping ratios. The smallest x values are limiting experiments. Disk gravity, gap structure, mass growth, and ordinary migration torques are absent.

5.2 Matched-target direct integrations

We test the selected case-24 architecture with full Newtonian integration between symmetric eccentricity kicks. A baseline kick preserves the osculating semimajor axis, orientation, and mean anomaly, and multiplies eccentricity by $\exp(\gamma\Delta t/2)$ on either side of the gravitational interval. Coordinates are synchronized and marked as modified after each kick. Inner eccentricities start at zero and the giant at 0.001. Primary integrations use $P_1/40$ and 20-yr kick intervals.

Eight fixed-duration pilot runs produced slightly different final giant eccentricities because short-period terms matter at such small initial e_g. We retain them as pilots. The primary eight runs instead stop pumping at the same target $e_g = 0.409649$, then apply immediate or gradual damping removal and follow 100,000 yr of gas-free evolution. Two mean-anomaly seeds are used per prescription. The endpoint is the same close-approach diagnostic as in the controlled experiment.

Slow pumping lasts approximately 35,500–36,200 yr. In the gradual-removal case, $\tau_d = 6013$ yr and damping decays for about 72,158 yr before the gas-free follow-up. The final giant eccentricity remains near 0.408–0.410. Slow growth without inner damping gives $\mathcal{F} = 0.0217$–0.0226, whereas gradual damping removal gives 0.0118–0.0153. Immediate removal with $d = 10$ leaves 0.369–0.410. All six complete.

At the primary resolution, the two $x = 1, d = 0$ cases encounter approximately 38,200 and 92,600 yr after pumping stops. However, both complete on refinement to $P_1/80$ and 5-yr kicks, with final $\mathcal{F} = 0.359$ and 0.490. Their higher excitation is reproduced, but the faster-growth survival disadvantage is not. Figure 4 retains this failed convergence check. Three refined slow histories also complete and preserve the action contrast (Table 2). Two seeds and one selected architecture cannot yield a general survival probability.

5.3 Damping with associated orbital drift

To test the fixed-semimajor-axis assumption, four further primary histories let damping conserve specific orbital angular momentum:

$$a' = a\frac{1-e^2}{1-e'^2}. \tag{10}$$

This prescription includes damping-associated inward drift, but not the full torque-driven migration of a gas disk. In one gradual-removal realization the inner axes shrink by about 12%, 20%, and 29%. Both gradual-removal cases complete, with $\mathcal{F} = 0.0662$–0.0930, while one of two instantaneous-removal cases encounters.

Two refined drift runs retain that particular encounter/completion contrast for seed 101. However, the gradual case changes from $\mathcal{F} = 0.0662$ to 0.0216. The action amplitude is not numerically converged, and this small experiment does not establish a sharp gas-removal threshold. It shows that low free action can arise without inserting the final coherent state, while identifying orbital drift as an important sensitivity.

6 Giant scattering

Giant scattering supplies a more self-consistent source of excitation, although our initial architecture grids re-

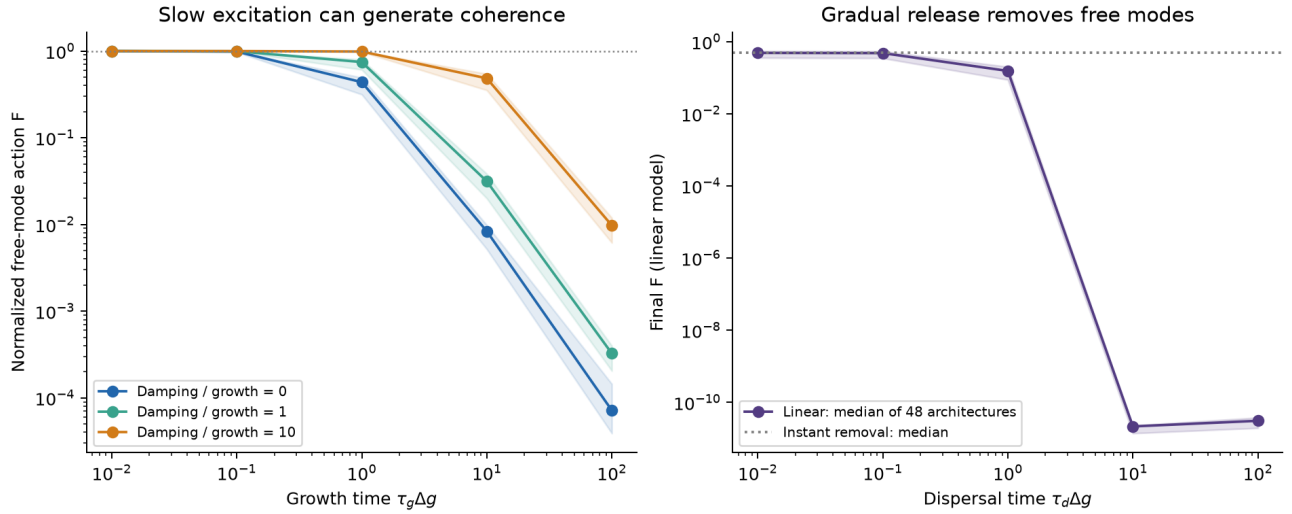


Figure 3: Generation of low free-mode action in 1,008 reduced linear histories. Shading represents the 16th–84th percentile across the 48 selected architectures, not uncertainty on a population mean. Left: giant eccentricity growth at different inner damping strengths. Right: removal of damping from the $x = 10, d = 10$ endpoints. The very small linear values do not carry over quantitatively to finite-eccentricity direct integrations.

Table 2: Matched-target growth histories in case 24. $\mathcal{F}$ is the median over the final 10,000 yr, sampled from osculating elements, when a run completes. The refined fixed-axis tests use $P_1/80$ and 5-yr kicks. The fast-growth survival contrast fails refinement; no survival threshold is inferred.

Prescription	Primary completions	Primary final $\mathcal{F}$	Refinement result
$x = 1, d = 0$, fixed axes	0/2	Encounter endpoints	Both complete; $\mathcal{F} = 0.359, 0.490$
$x = 10, d = 0$, fixed axes	2/2	0.0217–0.0226	One checked: 0.0218, complete
$x = 10, d = 10$, instant removal, fixed axes	2/2	0.369–0.410	One checked: 0.386, complete
$x = 10, d = 10$, $y = 10$, fixed axes	2/2	0.0118–0.0153	One checked: 0.0148, complete
$x = 10, d = 10$, instant removal, orbital drift	1/2	Survivor: 0.285	Encounter realization still encounters
$x = 10, d = 10$, $y = 10$, orbital drift	2/2	0.0662–0.0930	One checked: 0.0216, complete

main deliberately small. We retain the three case-24 inner planets at initial $e = 0.001$ and initialize outer giants at $e = 0.01$, inclination 0.5°, with randomized orientations and anomalies. Histories are integrated for up to 100,000 yr. An inner encounter terminates a run; it is not followed to physical destruction.

6.1 Two-giant instabilities

The two-giant grid uses masses $(3, 1), (1, 3), (3, 3)\, M_J$, the innermost giant at 3 au, and separation 2.5 mutual Hill radii. Four seeds give 12 primary histories with MERCURIUS [23] at $P_1/40$. Giant collisions merge perfectly inelastically, taking each initial radius as one Jupiter radius. Unbound bodies are removed only beyond 100 au, and energy changes from mergers and removals are accounted for explicitly. The artificial inner collision radii implement the encounter diagnostic, not physical planet radii.

Ten histories reach the horizon after a giant merger with the inner system retained, one ejects a giant and retains the inner system, and one reaches an inner encounter. The merger remnants have $e_g \simeq 0.004$–0.035 and $q_g/a_{i,\mathrm{out}} \simeq 6.9$–$8.1$. Their coexistence is largely explained by weak forcing; it does not require special alignment or low normalized $\mathcal{F}$. Near-zero giant eccentricity can make that normalization particularly misleading.

The ejection survivor has $e_g \simeq 0.232$, $q_g/a_{i,\mathrm{out}} \simeq 4.62$, and $\mathcal{F} \simeq 0.129$. A twice-finer realization also ejects a giant and retains the inner system but gives $\mathcal{F} \simeq 0.562$. The individual low-action amplitude is therefore not a converged prediction. Three refined two-giant runs all complete; the original encounter, which had an elevated energy residual, becomes a merger survivor. We do not count that coarse encounter as a verified physical failure.

6.2 Three giants: numerical sensitivity

The three-giant grid begins at 2 au, has 3.5 mutual-Hill-radius adjacent spacing, and uses equal masses of either

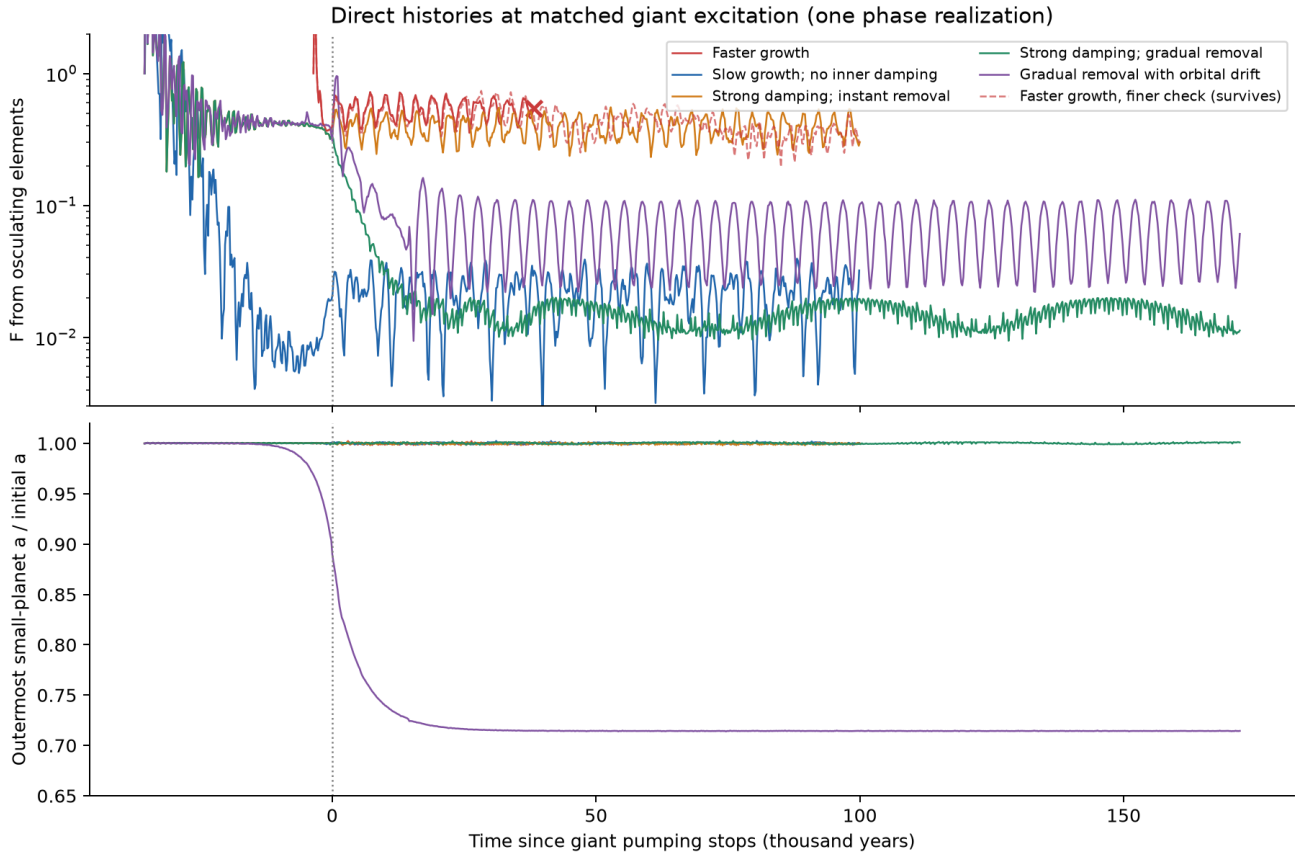


Figure 4: Generated histories in the selected compact architecture. Time zero marks the end of giant pumping. All histories begin with circular inner orbits. The cross marks the primary faster-growth encounter; the refined faster-growth trajectory completes. Green and purple curves have gradual damping removal; purple includes damping-associated orbital drift and has a resolution-sensitive final action. These differences constrain interpretation of the reduced disk model.

1 or 3 M_J. Eight MERCURIUS runs yield seven inner encounter endpoints and one surviving three-giant system. Eight reruns with TRACE [24], twice-finer timesteps, and adaptive stellar-pericenter handling reproduce the endpoint count but not the encounter times. Two TRACE encounters retain elevated energy residuals.

Independent IAS15 checks of those two realizations give one encounter at 17,960 yr with relative energy residual 2.3×10^{-10}, and one completion of its shorter 30,000-yr check horizon with residual 5.3×10^{-15}. Thus the seven-of-eight encounter count is not a converged physical fraction. The two- and three-giant grids also differ in initial radii, spacing, and masses, so their contrast cannot be attributed to planet number alone.

Scattering can leave weakly perturbed coexistence systems, produce relatively coherent survivors, or damage the inner system, consistent with history-dependent theory [12, 16]. We do not infer the frequencies of these channels. Planar free-mode diagnostics are omitted for contact-censored pilots and unsuitable highly inclined or multiple-remnant endpoints. In a chaotic problem, agreement of one trajectory is not a substitute for statistical convergence of a much larger ensemble.

7 Public survey tests

7.1 Fixed sample and injection recovery data

We freeze the California Legacy Survey (CLS) public repository [25, 26] at commit `689d491` (full hash in Appendix B). The planet table includes 164 known planets and 14 newly catalogued candidates under status codes K and C; both are retained. The primary cuts are inner minimum masses 2–30 $M_\oplus$ at 0.03–1 au and giant minimum masses 0.3–13 M_J at 1–10 au. This survey definition differs deliberately from the catalog's contextual union definition. Membership is determined from tabulated medians, without treating marginal uncertainty as independent posterior draws.

The selection gives 39 small planets, 68 giants around 58 giant hosts, and four coexistence hosts: HD 11964 A, HD 164922, HD 190360, and HD 219134. For a phase comparison we pair the outermost qualifying small planet with the innermost qualifying giant once per host. The cuts and one-pair-per-host rule were recorded before inspecting these conditional results; subsequent sensitivity analyses are exploratory, not externally preregistered

tests.

We retain all 58 per-star recovery files, totaling 58,000 injection tests, with pinned URLs, retrieval times, and hashes. The public snapshot supplies 1000 tests per star, compared with 3000 described in CLS I. The period floor is about two days. A secondary completeness benchmark therefore uses 0.04–1 au with the same mass interval, avoiding extrapolation below injection support. This common domain contains 6,432 injections and eight detected small planets in the same four hosts. The original primary cuts are unchanged.

The frozen repository provides marginal orbital summaries and author-side paths to chains, but does not supply the joint posterior chains for the four systems. We do not synthesize pseudo-posteriors from independently sampled marginal intervals. Unknown inclinations, node angles, and masses remain relevant to a three-dimensional dynamical interpretation.

7.2 Precision of eccentricity vectors

For the selected small planets, the quoted marginal 68% argument-of-periastron widths are approximately 122°, 97°, 22°, and 80°; for their giants they are 44°, 144°, 3°, and 127°. Only HD 190360 has both arguments tightly constrained. These RV arguments do not give mutual three-dimensional apsidal directions without inclinations and nodes. Circular-orbit angle degeneracy, wrapping, and posterior correlations preclude a valid alignment significance from four marginal point pairs.

We instead calculate a feasibility forecast: the inner eccentricity expected if the giant-dominated linear mode were occupied. This uses all qualifying planets in each host, minimum masses, the tabulated stellar mass, coplanarity, and leading relativistic precession $3GM_*n/(c^2a)$. The leading small-eccentricity RV harmonic is $K_i e_{i,\mathrm{coh}}$, where K_i is the tabulated semiamplitude. These are model predictions at point estimates, not inferred mode occupations.

The predicted harmonics span approximately 10^{-4}–2×10^{-2} m s^{-1} (Table 3). Even an ideal uniformly sampled sinusoidal fit with independent errors $\sigma_{\mathrm{RV}} = 1$ m s^{-1} would require

$$N_{3\sigma} \simeq \frac{18\sigma_{\mathrm{RV}}^2}{(K_i e_{i,\mathrm{coh}})^2}, \tag{11}$$

from $\sigma_A \simeq \sqrt{2/N}\,\sigma_{\mathrm{RV}}$. The resulting range is 4.4×10^4–1.8×10^9 measurements. Correlated noise and uncertain orbital parameters make this estimate optimistic. It identifies a poor coherence-test sample, not a practical observing proposal.

HD 190360's measured inner eccentricity is much larger than the pure giant-mode forecast. This argues against that particular mode occupation at the point estimates, but not against coherence being protective in a closely perturbed compact system. Its wide giant pericenter does not require the protection being tested in case 24.

Among 1,314 recovered common-domain injections across all giant hosts, mean $\cos(\omega_{\mathrm{inj}} - \omega_g) = 0.0055$ and half are within 90° of the giant argument. Among 229 recovered injections in the four coexistence hosts, the aligned fraction is 0.441. The files do not reveal a strong universal alignment preference in discovery recovery. They do not measure the distinct selection for an *informative eccentricity vector*, so this audit cannot calibrate a phase test.

7.3 Conditional completeness

For a transparent secondary calculation, we assume uniform planet density in $\log a$ and $\log m$ over the common domain. We estimate per-star recovery probability with a Gaussian kernel in those coordinates, using reference bandwidth 0.25 dex and a fixed 33×29 grid. Bandwidths of 0.15 and 0.4 dex and direct in-domain recovery fractions are retained as sensitivity checks. The injection eccentricities follow the survey prescription. We condition throughout on stars with *detected* qualifying giants.

We divide the 58 hosts at the median present tidal proxy

$$T_g = \sum_g \frac{m_g}{a_g^3(1-e_g^2)^{3/2}}, \tag{12}$$

which is $0.100924\,M_{\mathrm{J}}\,\mathrm{au}^{-3}$. This proxy is not a substitute for the full differential secular forcing. All four coexistence hosts, including all eight common-domain small planets, lie in the lower half. Summed mean completeness gives effective exposures of 7.864 and 5.600 stars in the lower and higher halves, respectively, each containing 29 actual stars.

The formal count/exposure estimates are 1.017 and 0 planets per detected-giant host, or 0.594 overall. The total varies from 0.554 to 0.670 across completeness estimators. These are planet-count benchmarks under an assumed distribution, not fractions of hosts containing planets. Zero detections also do not imply a zero upper uncertainty bound. Treating the eight planets as eight independent formation outcomes would overstate evidence because they occupy only four hosts.

An illustrative Bernoulli host model with assumed multiplicity from one to four planets per occupied host gives conditional probabilities of 0.088–0.106 that all four detected hosts lie in the lower half. These are plug-in, model-dependent tail calculations, not calibrated discovery p-values; host-fraction uncertainty and realistic within-system correlations are omitted. Median metallicities of the two halves are 0.192 and 0.160 dex. Four coexistence hosts cannot support a controlled analysis of metallicity, giant detection, architecture, and small-planet sensitivity.

The pattern is compatible with reduced forcing favoring coexistence, but does not establish a new empirical relation. Giant mass and separation effects have already been examined in CLS [3, 5]; stable-zone proxies also appear in the recent homogeneous GAPS analysis [9]. The

Table 3: The four selected CLS coexistence hosts. Separation uses the paired giant and outermost selected small planet. Coherent amplitudes include leading relativistic precession. The last column is the modeled harmonic, not the harmonic implied by the measured eccentricity.

Host	$q_g/a_{i,\mathrm{out}}$	Median measured e_i	Modeled $e_{i,\mathrm{coh}}$	K_i (m s^{-1})	$K_i e_{i,\mathrm{coh}}$ (m s^{-1})
HD 11964 A	12.36	0.106	0.004760	4.260	0.02028
HD 164922	5.89	0.096	0.000931	2.307	0.002147
HD 190360	20.56	0.165	0.001932	5.446	0.01052
HD 219134	12.34	0.077	0.000029	3.432	0.000101

public data chiefly identify where a test of orbital coherence lacks precision and dynamical relevance.

8 Conditions for coexistence

8.1 Three physical requirements

The combined evidence supports three conditional requirements. First, the formation history must supply and assemble the inner planets. The timing of giant growth relative to inner growth, material transport, and disk evolution matters [1]. Our simulations begin after planetary masses have been assigned, so this remains a literature-based prerequisite rather than a result of the integrations.

Second, the architecture and evolving perturbations must avoid destructive differential excitation. A sufficiently distant or low-eccentricity giant can leave the inner system weakly forced. A tightly coupled inner system or rapid additional apsidal precession can reduce its response. Secular resonances, close encounters, inclined perturbations, and other companions can undermine those protections. The large pericenter ratios in most catalog examples and the low-eccentricity scattering merger remnants illustrate coexistence without requiring a special apsidal state.

Third, where the giant's forcing is stronger, the inner eccentricity vectors can materially change the finite-time outcome. Small free components relative to an appropriate coherent mode can reduce relative orbital excursions. The matched circular/coherent/shuffled experiments isolate this effect, and selected long extensions show persistence beyond the main horizon. However, coherent preparation is neither necessary in weakly forced systems nor sufficient for indefinite survival. The operative requirement is avoiding damaging relative motion, not minimizing each planet's scalar eccentricity.

8.2 History and a testable prediction

The reduced disk calculations show that initially circular inner planets can reach low free action through sufficiently slow giant excitation or gradual removal of damping. Strong damping alone need not prepare the gas-free mode. Scattering can also leave benign remnants, but the final giant orbit does not uniquely encode what the inner system experienced.

A candidate observational prediction is therefore conditional: *at fixed, compact, approximately coplanar architecture, systems near a strong differential-forcing boundary should preferentially retain small free eccentricity vectors; this preference should weaken in widely separated, weakly forced systems.* The predicted quantity includes amplitudes and orientations together. Alignment alone does not identify gradual growth: in the single-frequency impulsive limit, $z_i/z_{\mathrm{forced}} = 1 - \exp(\mathrm{i}\Delta g t)$ has nonnegative real part despite substantial free eccentricity [16].

In the fixed-axis case-24 histories, gradual versus immediate damping removal reduces final $\mathcal{F}$ by roughly a factor of 25–35. That factor is not a population prediction: the orbital-drift experiment is quantitatively sensitive, physical disk-induced precession is absent, and only one architecture has direct growth histories. A credible quantitative test would need a broad physical disk ensemble and an observational likelihood for full eccentricity vectors, including their measurement selection. The present CLS sample supplies neither a strongly forced comparison set nor sufficiently informative vectors.

The contribution of this work is the explicit controlled comparison, the connected history tests, and the reproducible survey-feasibility audit. Slow-excitation coherence, forced-eccentricity protection, and scattering-history dependence already have direct precedents [15, 10, 12, 13, 16]. We do not claim a new secular mechanism or an independently established novel population law.

8.3 Numerical and physical scope

The direct disk integrations omit disk gravity, full migration torques, planet mass growth, inclination evolution, stellar oblateness, tides, and relativistic terms. The observed-system mode forecasts include leading relativity but remain linear, coplanar, and based on minimum masses. In case 24, relativistic contributions are below 0.6% of the corresponding Newtonian diagonal rates; even a small frequency shift can matter near a resonance, so this comparison does not establish relativistic robustness.

The stress selection, two phase seeds per architecture, single direct-growth architecture, small scattering grids, and finite integration horizons prevent population inference. Coarse-to-fine endpoint changes are retained. In particular, the faster-growth encounter contrast disappears entirely on refinement, and one low-action scattering realization does not preserve its amplitude. No sim-

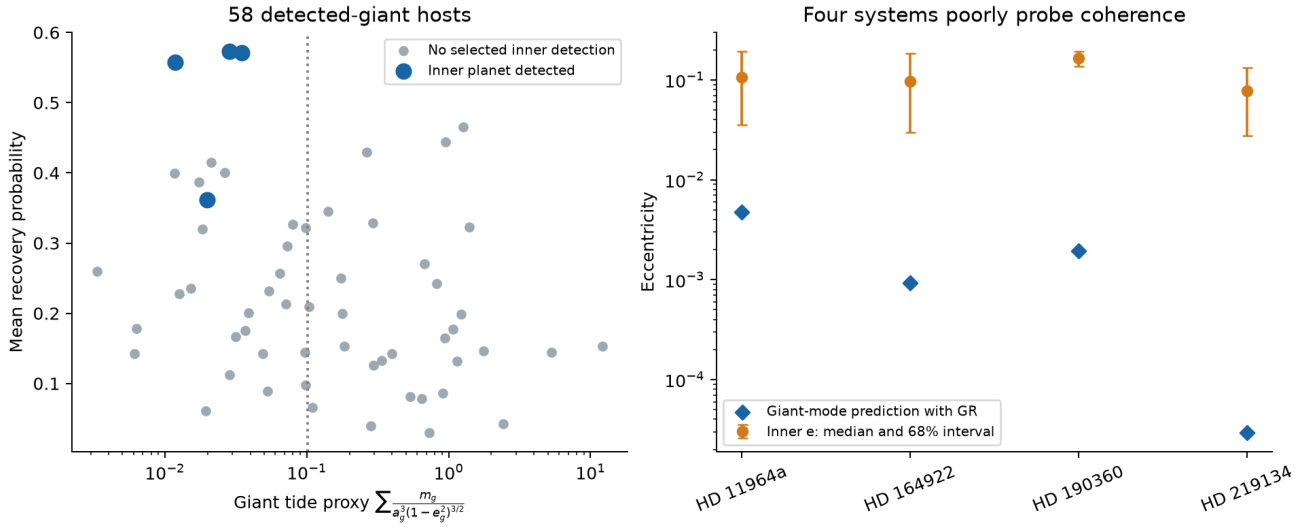


Figure 5: Public-survey feasibility. Left: common-domain recovery probability versus present tidal proxy. The dotted line is the median split; blue symbols identify the four detected coexistence hosts. Right: measured marginal inner eccentricities with 68% intervals and model coherent amplitudes. These comparatively wide systems need not occupy a pure giant mode to coexist. The figure is a selection and measurement audit, not a posterior test of mode occupation.

ulation here establishes gigayear stability or the physical destruction of a planet at our encounter endpoint.

9 Conclusions

Outer giants can coexist with inner small planets when their formation histories produce both populations and subsequent evolution leaves sufficiently little damaging differential excitation in the inner system. Weak external forcing, strong internal coupling or additional precession, and coherent eccentricity vectors are complementary ways to satisfy the dynamical part of this condition.

Our controlled integrations demonstrate that orbital state can change finite-time encounter outcomes at fixed giant mass, semimajor axis, and eccentricity. Reduced growth and damping-removal models can generate low free action from circular starts, but a general survival boundary has not been established. Scattering admits both weakly forced survivors and damaging histories. The selected public survey systems mainly test weak forcing and have insufficiently informative eccentricity vectors to distinguish the proposed coherence mechanism.

The evidence therefore favors a conditional answer involving architecture, orbital state, and excitation history. It does not justify a universal giant-property threshold, compulsory apsidal alignment, a uniquely identified formation channel, or a new occurrence law.

Data and code availability

The retained research products include public catalog and survey snapshots, queries, pinned source URLs and hashes, architecture definitions, random seeds, direct-run outcomes and sampled histories, numerical checks, and figure scripts. The two accompanying research archives preserve the baseline and history/survey analyses separately. The typesetting source for this combined manuscript includes the figures and can be compiled independently; it does not replace the full numerical archives. No public repository DOI or preprint identifier has yet been assigned.

Authorship and preparation

Author names and affiliations are placeholders. This manuscript was assembled with AI-assisted literature synthesis, code generation, numerical analysis, and writing. Scientific responsibility, authorship, and independent expert review remain to be established before submission. No preprint submission or peer-review acceptance is implied.

A Validation and run inventory

Independent quadrature checks the Laplace coefficients to maximum relative discrepancy 2.0×10^{-13} in the tested cases. Independent complex-ODE integration agrees with the mode solution to 2.7×10^{-13} in absolute eccentricity. The quadratic secular angular momentum deficit is conserved to relative 1.4×10^{-15}, and the conic-intersection formula agrees with direct polar sampling for 1000 random ellipse pairs. Increasing secular screen sampling from 4,096 to 16,384 epochs changes three pilot maxima by at most 0.015%.

For the growth problem, eigendecomposition agrees with an adaptive ODE solution to 1.3×10^{-11} and a sep-

Table 4: Inventory of distinct jobs. Numerical checks revisit initial conditions; an executed job may end at an encounter. The two sides list different experiment families, not matched pairs.

Controlled-state experiments	Jobs
Development pilots	6
Four-state primary comparison	384
Selected IAS15/finer WHFast checks	24
Additional full-cohort refinement	84
Loading/eccentricity scan	70
One-million-year extensions	12
No-giant controls	48
Controlled-state subtotal	628
Reduced linear histories	
Eccentricity growth	720
Damping removal	288
Linear-history total	1008

Generated-history experiments	Jobs
Fixed-duration growth pilots	8
Matched-target growth	8
Slow-growth refinement	3
Faster-growth refinement	2
Damping-associated drift	4
Drift refinement	2
Contact-censored scattering pilots	6
Two-giant merger/ejection histories	12
Two-giant refinement	3
Three-giant histories	8
Three-giant TRACE checks	8
Three-giant IAS15 checks	2
History subtotal	66

Total distinct direct integration jobs: $628 + 66 = 694$.

arate matrix-exponential calculation to 1.6×10^{-14}. The $\mathcal{F} = \eta^2$ identity and the damping angular-momentum identity are verified. A one-orbit kick test agrees with analytic exponential eccentricity change and a synchronized implementation to better than 10^{-10}.

Among completed primary controlled integrations, the largest sampled relative energy error is 9.4×10^{-8}. Regular prepared case-24 IAS15 runs have errors below 6×10^{-15}. Successful primary growth histories have residual gravitational energy errors of order 10^{-11} after subtracting applied work. Energy diagnostics check numerical implementation; they neither validate the disk model nor bound errors between sampled outputs near a close encounter.

Table 4 counts distinct jobs. The 96-run stress-cohort refinement combines 84 additional runs with 12 finer WHFast runs already in the selected 24-run validation set. They must not be counted twice. Six baseline pilots and six contact-censored scattering pilots remain in the archives. Contact-endpoint action estimates are excluded from the physical conclusions. The two IAS15 scattering checks stop at 30,000 yr unless an encounter occurs, rather than the 100,000-yr main scattering horizon.

B Survey provenance

The full frozen CLS repository revision is

`689d49178f8cab5a0c8d5291035786423c530953`.

The retained planet and stellar tables, observing summaries, 58 recovery files, and manifests make the selection auditable. Mass and axis columns are interpreted in Jupiter masses and au before conversion. Published lower/upper quantile columns are quantile values, not symmetric errors. Missing observing metadata for one selected host is retained as missing rather than filled with zero or used to drop that star.

The primary observational hypothesis concerned apsidal alignment among eccentric small planets and eccentric giants. The full-vector formulation arose during interpretation of the history experiments. The fixed analysis plan and follow-up decision log distinguish these stages. No external preregistration or independent validation sample is claimed.

The illustrative host-level calculation conditions on exactly four detected coexistence hosts and assumes identical multiplicity per occupied host. It is only a sensitivity check on the effect of counting planets instead of systems. A selection-aware population analysis would additionally need a latent multiplicity distribution, joint orbital posteriors, uncertainty in completeness and host occupation, giant detection selection, and stellar covariates. Those quantities cannot be identified from four detected hosts in this restricted sample.

Analysis scripts were replayed from the retained research archives without rerunning the direct integrations. Numerical summaries and scientific figures reproduced, with only negligible floating-point serialization differences in two extension tables. The new combined manuscript reorganizes those retained results; it adds no new dynamical integrations or independent observations.

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